THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 162	Let breadth $= b$.
2	**(b)** $\frac{23}{24}$	Use the LCM of the denominators.
3	**(c)** 2	
4	**(d)** 5	Number of sides $= 360 \div a$.
5	**(b)** 22	Simplify step by step.
6	**(c)** 16	Perimeters are equal.
7	**(b)** 60	Difference of the fractions, then divide.
8	**(a)** 20	Add the three sides.
9	**(b)** 45	Divide.
10	**(d)** 30	Opposites cancel.
11	**(d)** 120	Half of the rectangle.
12	**(b)** 40	Fraction removed = difference.
13	**(a)** Both A and R are true and R is the correct explanation of A.	2 x (12 + 5) = 34.
14	**(a)** Both A and R are true and R is the correct explanation of A.	Multiplying 3 and 4 by 2 gives 6 and 8.
15	**(a)** Both A and R are true and R is the correct explanation of A.	Every chord through the centre has length 2r, and no other chord is longer.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$l + b = 30$; ratio parts $= 5$ (1); Length $= 18$ cm, breadth $= 12$ cm (1).	1+1
17	Number $= 18 \div$ \frac{3}{4}$= 24$ (1); $\frac{1}{3}$ of the number $= 8$ (1).	1+1
18	Inside: 2; On: 1 (1); Outside: 1 (1).	1+1
19	Sides $= 360 \div 30 = 12$ (1); Perimeter $= 12 \times 4 = 48$ cm (1).	1+1
20	Opposite integers cancel each other; Remaining: integers from 7 to 10 (1); Sum $= 34$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Perimeter $= 130$ m; Wire needed $= 3 \times 130 = 390$ m (1); Fencing cost $= 4680$ (1); Ploughing cost $= 4000$ (1).	1+1+1
22	Girls $=$ \frac{5}{8}$\times 480 = 300$ (1); Chess girls $=$ \frac{2}{5}$\times 300 = 120$ (1); Boys $= 480 - 300 = 180$ (1).	1+1+1
23	Distance between adjacent centres $= 2r = 6$ cm (1); Side $= 6$ cm (1); Area $= 36$ cm^2 (1).	1+1+1
24	Angle $= 360 \div 5 = 72^\circ$ (1); Lines of symmetry $= 5$ (1); Total around centre $= 5 \times 72 = 360^\circ$ (1).	1+1+1
	OR: After dive: -80; After rise: -40 (1); After second dive: -65 (1); Total distance $= 115$ m (1).	1+1+1
25	Square perimeter $= 4 \times 12 = 48$ cm; rectangle perimeter $= 2(14 + 10) = 48$ cm; Square area $= 144$ cm^2 (1); Rectangle area $= 140$ cm^2 (1); Square is larger by 4 cm^2 (1).	1+1+1
	OR: $\frac{?}{12} =$ \frac{5}{6}$-$\frac{1}{4} (1); $= \frac{7}{12} =$ \frac{7}{12} (1); Missing numerator $= 7$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $2 \times (40 + 25) = 130$ m (1). (ii) $40 \times 25 = 1000$ m^2 (1). (iii) $130 \times 20 =$ ₹2600 (1). (iv) $1000 \div 50 = 20$ parts, so $20 \times 2 = 40$ kg (2).	1+1+1+2

27	Draw AB = 6 cm (1); With A as centre and radius 10 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 8 cm (1); Half diagonal OA $= 5$ cm (1); Perimeter $= 2(6 + 8) = 28$ cm (1).	1+1+1+1+1
	OR: (a) 6 (1); (b) 60° (1); (c) 30 cm (1); (d) 240° (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6

Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 336	Outer minus inner.
2	**(b)** 8	Divide the length by the piece length.
3	**(d)** 16	Two diameters.
4	**(d)** 150	Add the smallest angles.
5	**(a)** -17	Subtract.
6	**(d)** 14	Perimeter does not change.
7	**(c)** 5	Find the number first.
8	**(a)** 16	Each is half the diagonal.
9	**(d)** 3	If a figure has a line and order n, it has n lines.
10	**(c)** -30	Gains positive, losses negative.
11	**(d)** 5	Area = cost ÷ rate; breadth = area ÷ length.
12	**(c)** $\frac{7}{10}$	Compare with a common denominator.
13	**(a)** Both A and R are true and R is the correct explanation of A.	A square is a regular polygon with 4 sides.
14	**(a)** Both A and R are true and R is the correct explanation of A.	-7 lies to the left of -3.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	7 x 7 = 49 and area is in square units, but R does not explain the value 49.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Area $= 1200$ m^2, perimeter $= 140$ m; Levelling cost $= 6000$ (1); Fencing cost $= 1680$; total $= 7680$ (1).	1+1
17	Required =\frac{5}{6}−\frac{2}{7} (1); $= \frac{23}{42}$ (1).	1+1
18	$\angle AOB = \angle COD = 35^\circ$ (vertically opposite) (1); $\angle BOC = \angle AOD = 145^\circ$ (1).	1+1
19	Order $= 360 \div 40 = 9$ (1); 120° is a multiple: 3 times 40 (1).	1+1
20	$? = 5 - (-9)$ (1); $? = 14$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Width of ring $= 5$ cm (1); Inner diameter $= 8$ cm (1); Longest chord of outer circle = its diameter $= 18$ cm (1).	1+1+1
22	Lines through opposite vertices: $4 \div 2 = 2$ (1); Lines through midpoints of opposite sides: 2 (1); Total $= 4$ (1).	1+1+1
23	Other $= -4 - 11 = -15$ (1); Difference $= 26$ (1); Check the sum: -4 (1).	1+1+1
24	Floor in cm: 600 by 400; Tile area 600 cm^2 (1); Floor area 240000 cm^2 (1); Tiles $= 400$ (1).	1+1+1
	OR: Together =\frac{1}{4}+\frac{1}{6}= $\frac{5}{12}$ (1); Remaining $= 1 - \frac{5}{12} = \frac{7}{12}$ (1); State the answer (1).	1+1+1
25	BM $= 5$ cm, BN $= 3$ cm (1); Perimeter $= 2(5 + 3) = 16$ cm (1); Area $= 15$ cm^2 (1).	1+1+1
	OR: Smallest angle $= 360 \div 6 = 60^\circ$ (1); $180 \div 60 = 3$ rotations (1); A whole number means the figure fits (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$ (1). (ii) $1 - \frac{5}{8} = \frac{3}{8}$ (1). (iii) Kavita (3 slices against 2) (1). (iv) 3 slices are left (1); $\frac{3}{8} = \frac{6}{16}$ (1).	1+1+1+2
27	Sum of whole parts $= 6$ (1); Sum of fractions $= 1$ (1); Total $= 7$ (1); After subtracting: $6\frac{1}{4}$ (2).	1+1+1+2
	OR: Row length $= 5 \times 6 = 30$ cm (1); First to last centres $= (5 - 1) \times 6 = 24$ cm (1); Circles in the strip $= 7$ (1); Show the working with diameter (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 64	A square has the greatest area for a given perimeter.
2	**(c)** $2\frac{3}{4}$	Subtract.
3	**(c)** 70	Base angles are equal.
4	**(b)** 9	$360 \div$ angle.
5	**(b)** -5	Add b to both sides.
6	**(b)** 42	Add the three sides.
7	**(d)** 24	LCM.
8	**(b)** 10	Diameter equals the side.
9	**(b)** 42	Number of sides times side.
10	**(b)** 2	Subtracting a negative adds.
11	**(a)** 84	Outer side $= a + 2$.
12	**(c)** $1\frac{1}{6}$	Use the LCM of the denominators.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true. R is just another way of writing the number and does not explain why it is improper.
14	**(c)** A is true but R is false.	Radius = 14 / 2 = 7, so A is true. The diameter is twice the radius, so R is false.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true but the order of rotation does not explain the number of lines.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Room area $= 48$ m^2; Carpet area $= 20$ m^2 (1); Uncovered $= 28$ m^2 (1).	1+1
17	One way $= \frac{23}{20}$ km (1); Total $= 2\times = \frac{23}{10}$ km (1).	1+1
18	Area of square $= 100$ cm^2 (1); Area of AOB $= 25$ cm^2 (one quarter) (1).	1+1
19	Smallest angle $= 360 \div 4 = 90^\circ$ (1); Angles: 90°, 180°, 270° (1).	1+1
20	Final: $-20 + 60 - 25 = 15$ m (1); Distance from 30 m: 15 m (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Common difference $= 6$ (1); Next terms: 4, 10, 16 (1); First positive term: 4 (1).	1+1+1
22	Perimeter $= 160$ m; Wire needed $= 2 \times 160 = 320$ m (1); Fencing cost $= 4800$ (1); Ploughing cost $= 7500$ (1).	1+1+1
23	Girls =\frac{4}{9}$\times 360 = 160$ (1); Chess girls =\frac{3}{8}$\times 160 = 60$ (1); Boys $= 360 - 160 = 200$ (1).	1+1+1
24	Distance between adjacent centres $= 2r = 8$ cm (1); Side $= 8$ cm (1); Area $= 64$ cm^2 (1).	1+1+1
	OR: Angle $= 360 \div 6 = 60^\circ$ (1); Lines of symmetry $= 6$ (1); Total around centre $= 6 \times 60 = 360^\circ$ (1).	1+1+1
25	After dive: -80; After rise: -25 (1); After second dive: -55 (1); Total distance $= 155$ m (1).	1+1+1
	OR: Square perimeter $= 4 \times 15 = 60$ cm; rectangle perimeter $= 2(18 + 12) = 60$ cm; Square area $= 225$ cm^2 (1); Rectangle area $= 216$ cm^2 (1); Square is larger by 9 cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $2 \times 3 = 6$ cm (1). (ii) $8 - 5 = 3$ cm (1). (iii) The diameter, $2 \times 8 = 16$ cm (1). (iv) P is outside the 3 cm circle, on the 5 cm circle and inside the 8 cm circle (2).	1+1+1+2

27	Rectangle area $= 240$ cm^2 (1); Triangle area $= \frac{1}{2} \times 24 \times 6 = 72$ cm^2 (1); Remaining $= 168$ cm^2 (1); Fraction left $= \frac{168}{240}$ (2).	1+1+1+2
	OR: After day 1: \frac{3}{4} left (1); Day 2 sells $\frac{1}{2}$ of total (1); Left fraction $\frac{1}{4}$ (1); Total $= 12 \div \frac{1}{4} = 48$ m (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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